

Quantifying Routing-Induced Frequency Dispersion in an Open-Source SKY130 Ring-Oscillator PUF: A Pre-Silicon Study

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Abstract

A ring-oscillator physical unclonable function (RO-PUF) turns manufacturing variation between nominally identical oscillators into a device secret. The physical implementation can add a frequency pattern of its own on top of that variation. This paper quantifies one component of that pattern, the instance-to-instance routing capacitance assigned by the automated OpenLane/OpenROAD flow on the open SKY130 process design kit, using a reduced model: nominal transistor-level models of the verified post-route oscillator topology, loaded with each ring net's total extracted capacitance from the final SPEF. No fabricated devices have been measured yet, and distributed resistance and coupling are not represented.

Under this model, the 16 automatically placed oscillators of Arm A in the candidate build spread 5.53% peak to peak (population SD 1.65%, mean 554.7 MHz), with extracted ring capacitance running from 10.9 to 17.0 fF. Because a single build only shows one draw of the router, I repeated the whole flow nine times over a narrow band of placement density with the source, floorplan, constraints, tool, and PDK frozen. Dispersion across those nine builds has a median of 5.75% and a range of 4.19% to 6.99%. What does not move is the mechanism: every build gives a frequency-capacitance correlation of about -0.999, and the fitted slope of -4.94 MHz/fF in the candidate build matches the -4.93 MHz/fF obtained earlier from an independent 32-oscillator layout. That earlier fit also predicts this build's 16 individual frequencies to a mean absolute error near 0.1%. An older build reported 10.5%, well above the sweep, and it owed most of that range to one oscillator the router had loaded with 24.4 fF; no oscillator in the candidate build is loaded that way. So a peak-to-peak number describes the build it came from, and I report the nine builds together for that reason.

The comparison arm, Arm B, uses one hardened oscillator macro sixteen times. A single extraction of that macro gives a 569.5 MHz reference for the shared internal layout, which removes internal-layout variation by construction but does not on its own prove a smaller total spread once the chips are made.

The contribution is a pre-fabrication diagnostic, traceable in the repository, that separates nominal layout bias from the variation a PUF is supposed to use. Whether the layout pattern repeats across dies, reduces uniqueness, or enables a practical prediction attack is left for the silicon phase, not claimed here.

1. Introduction

An RO-PUF compares the frequencies of nominally identical ring oscillators and converts comparisons into response bits [1]. Ideally the useful chip-to-chip differences come from manufacturing variation. In practice, placement and routing give instances different parasitic loads, which adds a deterministic component to the comparison.

A fixed layout pattern is not automatically a security failure. Random mismatch may be larger, comparisons may cancel shared structure, and response processing may absorb bias. But a layout component that repeats across chips could reduce uniqueness or make some comparisons predictable, and deciding between those outcomes takes measurements from multiple fabricated devices.

This paper reports the pre-silicon part of that investigation. The question is narrow: with transistor parameters held fixed, how much frequency variation does the physical implementation of one open-source RO-PUF design introduce? I contribute nominal post-layout results for two automatically placed oscillator arrays, a capacitance-based explanation for the observed spread, a matched-macro arm that gives every instance the same internal geometry, the scripts to run the same diagnostic before any fabrication, and the specific predictions I plan to test once chips come back.

2. Background and related work

Suh and Devadas introduced the widely used RO-PUF construction [1]. Maiti and Schaumont examined improved ring-oscillator PUF designs and compensation for systematic effects [2], and later FPGA studies mapped spatial variation and placement-dependent behaviour [3, 5]. Other work proposes statistical bias reduction, configurable structures, and placement-aware designs [5-8]. Katzenbeisser et al. evaluated several PUF constructions, including ring oscillators, across 96 ASICs [12].

The literature does not support a simple rule that any systematic structure makes an RO-PUF predictable. Wilde, Hiller, and Pehl found that adjacent-oscillator comparisons reduced exploitable spatial structure in their data, and that the estimated covariance was too small for their predictor to beat the relevant baseline [4]. That is important counterevidence: layout bias has to be judged together with the comparison scheme, the mismatch distribution, and the attacker model.

Open-source ASIC flows allow an experiment that closed flows make awkward: the designer can read the routed netlist and the parasitic extraction before fabrication instead of treating the implementation as opaque. OpenLane/OpenROAD [9] and the open SKY130 PDK [10] provide that setting, and a TinyTapeout RO-PUF project shows the circuit family works in the same ecosystem [11]. What I did not find is the measurement this paper makes: the nominal frequency component tied to instance-dependent routing in one automated ASIC layout, quantified from the flow's own extraction.

3. Design under test

The candidate design, tt_um_nikodemetrashvili20_ro_puf, occupies a TinyTapeout 2x2 tile. It holds two 16-oscillator arms and a shared serial measurement core that enables one oscillator at a time and counts its edges over a fixed window.

Each oscillator is a 31-stage ring of SKY130 standard cells: an enable NAND, 30 inverters, and an isolating output buffer tapped near the middle of the chain. A nominal pre-layout SPICE control sits near 633 MHz. Arm A lets the flow place and route each oscillator with the surrounding logic. Arm B places 16 copies of one hardened oscillator macro on a regular grid. The logical circuit is identical in both arms; the physical implementation method is the experimental variable. Figure 1 summarizes the design.

One design, two layout methods. Nominal post-layout simulation isolates the implementation contribution; silicon adds variation.

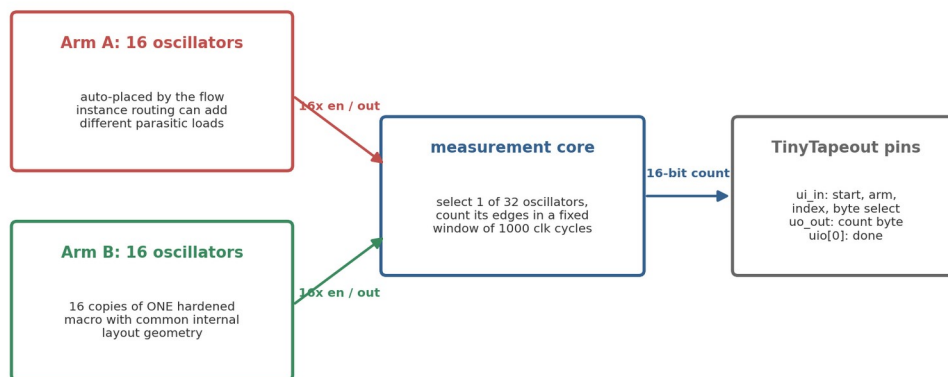


Figure 1. Block diagram of the two-arm design. Arm A lets the flow place and route each oscillator separately; Arm B repeats one hardened macro with a common internal layout.

How the measurement window is closed turned out to matter more than I first assumed. An earlier revision clocked the ripple counter through an AND gate that combined the selected oscillator with the window enable. That enable is generated in the reference-clock domain and can fall at any point in a 570 MHz cycle, so the gate could cut the final pulse on the counter's clock net down to a sliver, which is the classic way to put a flip-flop into an illegal state. The current design removes the gate. The counter is clocked straight from the selected ring, and the window is enforced by enabling and disabling the ring itself through its own NAND stage.

That change moves the boundary problem rather than removing it. Whenever the enable drops, the last pulse the ring produces can be almost any width depending on the phase, and a sweep of the extracted oscillator shows this directly: at some phases the final pulse falls to about 175 ps against a nominal half-period of 846 ps. The question is whether the counter minds. To find out I drove a real SKY130 dfrtp flip-flop, wired as the first ripple stage, from the extracted ring and dropped the enable at 38 phases covering a full period.

At every phase the flop settled to a clean rail, never sat near mid-supply, and the captured total varied by exactly one count between phases. The ring stop therefore costs at most one edge in roughly twenty thousand, which the core's settle handshake absorbs by publishing a count only after three consecutive equal reads. Decks and checkers are in `sim/spice/gono/`.

One caveat matters for interpretation: the two arms are not identical apart from internal-layout matching. Hardening inserts input and output delay buffers at the Arm B macro boundary that Arm A does not have, the sixteen Arm B macros occupy a regular block on one side of the tile while Arm A fills the rest, and the two arms differ in local decap and power-delivery geometry. The comparison is therefore a matched hardened-macro implementation against a conventional automated standard-cell implementation, not two circuits that differ only in internal routing; the ring loops themselves stay logically equivalent.

The boundary difference is larger than the buffer count suggests, and worth quantifying rather than mentioning. Taken from the top-level extraction, the capacitance each arm's ring output drives spans 0.24 to 2.71 fF in Arm A, mean 0.84, against 2.89 to 29.46 fF in Arm B, mean 14.46. Arm B drives roughly seventeen times the load, because its macros sit at fixed positions across the tile while Arm A's oscillators are placed near the shared multiplexer. That load is also why the flow puts a driver on the macro output; attempting to remove it only moves the problem, since the resizer then alters the ring's tap buffer instead.

The same geometry places the two arms in different parts of the die. The sixteen macros tile a region roughly 300 by 184 micrometres, while the automatically placed oscillators occupy a box of about 44 by 78 on one side. I tested whether a less one-sided floorplan could remove this, by skipping the middle column of the power grid so a standard-cell channel runs through the macro field. Two variants built and passed every physical check, and both spread Arm A considerably wider, but in each one the placer stretched a single oscillator into a thin line, 126 and 106 micrometres long, roughly doubling that instance's routing load and pushing the array's capacitance spread from 44% of the mean to 199% and 135%. Four macro rows is the maximum that fits the die, sixteen macros in four rows therefore need four columns, and only five column positions align with the power grid, so the shipped arrangement is the one that leaves the automatically placed arm a single contiguous region. The regional difference is a consequence of die area, macro footprint and grid pitch rather than a tuning choice, and removing it would need a larger tile. Both trials are recorded in the repository.

What protects the frequency result is that all of this lies outside the oscillator loop. The loop is the enable NAND and thirty inverters with feedback, and the only boundary cell touching it is the tap buffer, which is a `buf_1` in both arms. The enable buffer is further irrelevant during a measurement, since the enable is static while the window is open and the ring is driven through the feedback node. So the extra output stage and the heavier output route cannot bias the frequencies reported here, though they would have to be accounted for in any comparison of edge quality or bit reliability between the arms.

The main results below come from a coherent build of the current RTL that passes the physical checks (Magic and KLayout DRC, XOR, LVS, antenna, detailed route, power grid) with zero violations; two earlier builds, an archived dual-arm snapshot and a 32-oscillator layout, are reported for contrast (see SIGNOFF.md in the repository).

4. Method

The analysis is four separate operations, each traceable to a checked-in file. First, the physical flow produces the gate-level netlist, DEF, and nominal-corner SPEF. Second, a structural verifier (`verify_ring_topology.py`) parses the final netlist and confirms that every Arm A oscillator kept exactly one enable NAND, 30 inverters, and one tap buffer, with no cell inserted into the loop; the deck generation is only valid if this holds. Third, the generator reconstructs that verified topology from nominal SKY130 transistor-level cell models and attaches each ring net's SPEF `*D_NET` total capacitance to the matching node as a grounded lumped capacitor. It does not simulate the extracted network itself. Fourth, ngspice transient simulation (1.8 V, 27 C, nominal TT models) measures each oscillator's frequency, with a parallel control deck carrying no extracted capacitance. The SPEF declares `PIN_CAP NONE`, so pin capacitance is not part of the transferred load.

On the chip, one oscillator runs at a time. The combined simulation deck enables all 16 reconstructed oscillators concurrently, which is equivalent under this model because they share only an ideal supply and the reduced model carries no coupling between them. Distributed wire resistance is omitted entirely; its effect is not quantified here and is one reason the distributed-RC comparison is required. Each oscillator starts disabled and is released by an enable pulse, which avoids a simulation artefact where an artificial initial condition excited an unintended ring mode. Frequency is measured over 20 periods after startup.

For the matched arm, the hardened macro is extracted and simulated once. The 16 instances share one internal GDS, so this single result serves as the internal-layout reference for all of them.

For each automatically placed array I report mean, standard deviation, range, peak-to-peak spread relative to the mean, and the Pearson correlation between frequency and extracted ring capacitance. Instances inside one routed design are not independent samples from a chip population, so these are descriptive statistics for that layout. Standard deviations are population values across those instances, which are the complete set for a layout, not a sample from a wider population. The extracted capacitance is also the only spread-producing input to the model, so a strong frequency-capacitance correlation is expected almost by construction. The coefficient confirms the model behaves; the spread it produces, and the slope, are the physically interesting parts.

5. Automatically placed arrays

5.1 Earlier 32-oscillator layout

The earlier build placed all 32 oscillators automatically. In the no-parasitic control every deck produced the same nominal frequency, approximately 633.640 MHz, which checks the

generator itself. With extracted capacitance the mean was 567.6 MHz with a population standard deviation of 10.8 MHz (1.90%). Frequencies ranged from about 539 to 589 MHz, a 50.2 MHz or 8.8% peak-to-peak spread.

Extracted ring capacitance ranged from 7.4 to 17.8 fF. Frequency and capacitance had Pearson $r = -0.997$ with a fitted slope of about -4.93 MHz/fF (Figure 2a). Correlations with placement coordinates were much weaker ($|r| < 0.27$; Figure 2b). For this layout the spread behaves like an instance-specific routing load, not a smooth die-wide gradient.

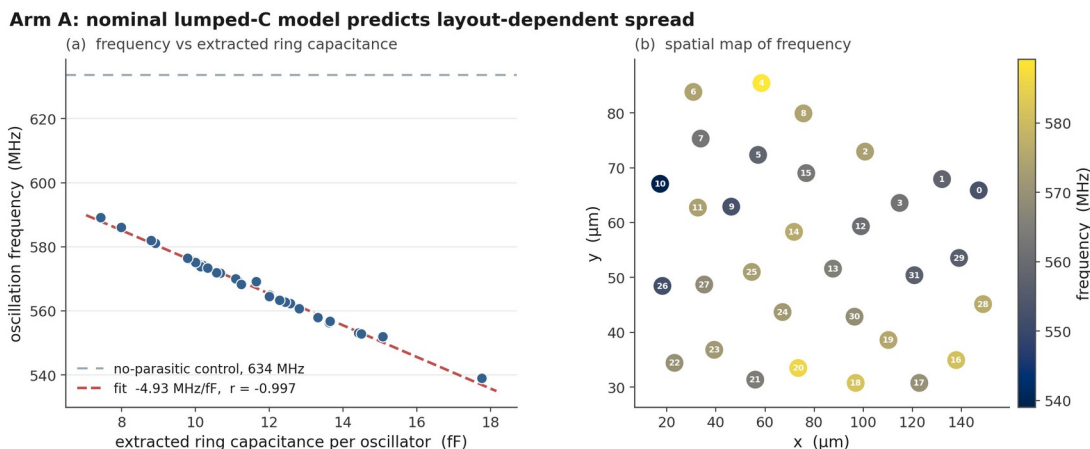


Figure 2. Nominal post-layout results for the earlier 32-oscillator layout: frequency versus extracted ring capacitance with the no-parasitic control, and a spatial frequency map.

5.2 Coherent dual-arm build

Arm A of the candidate build contains 16 automatically placed oscillators. Their nominal post-layout frequencies average 554.7 MHz and span 30.7 MHz from end to end, which is 5.53% of the mean, with a population SD of 9.15 MHz (1.65%). The slowest ring is R014 at 540.0 MHz carrying 17.0 fF; the fastest is R07 at 570.7 MHz carrying 10.9 fF. Correlation against extracted ring capacitance is -0.9997 and the fitted slope is -4.94 MHz/fF (Figure 4). The loads in this build form a fairly smooth band rather than a distribution with a straggler: the two heaviest rings differ by 0.34 fF, so no single instance dominates the range, and dropping any one oscillator leaves the spread between 4.7% and 5.5%. The build passes Magic DRC, KLayout DRC, XOR, LVS, antenna, and power grid with zero violations, and the netlist verifier confirms all 16 rings survived place and route with no cell inserted into a loop, so the dispersion estimate and the candidate GDS come from the same signed-off run.

An earlier build of this design reported 10.5%. That number came from a layout in which the router gave one oscillator 24.35 fF while the other fifteen sat between 11.7 and 16.95 fF, and that single instance carried most of the extreme range; without it the same build measured 4.60%. Nothing was wrong with the build, and the heavy instance was a real routing outcome rather than a measurement fault, but a peak-to-peak statistic computed over sixteen instances is sensitive to exactly that kind of draw. The sweep in Section 5.3 is what settles how typical it was.

Putting two builds' spread figures side by side is a fairly weak comparison. A better one is whether a fit from one build predicts the individual oscillators of another. A linear capacitance fit trained only on the 32-oscillator build (624.6 MHz - 4.93 MHz/fF) predicts the frequencies of later builds to roughly 0.1% mean absolute error with rank correlation near 0.997, and the candidate build's own fitted slope, -4.94 MHz/fF, sits within a fraction of a percent of that independently trained value. Each layout gets its own pattern of loads. The relationship converting those loads into frequencies is the same one every time.

5.3 Dispersion across nine builds

A peak-to-peak figure from one layout says little about what the flow does in general. LibreLane 3.0.3 does not expose the place-and-route seed, so a strict seed-only replicate set was not available. Instead I varied one neutral placement knob, the target placement density, in 1% steps from 56% to 64%, and froze everything else: the RTL, the macro locations, the floorplan, the constraints, the tool version, and the PDK. All nine builds hardened, and all nine kept every ring intact. Read the result as a placement-sensitivity band rather than a seed distribution.

Dispersion across the nine builds has a median of 5.75%, a range of 4.19% to 6.99%, and a standard deviation of 0.80% (Figure 5). The candidate build sits at 5.53%, close to the middle. Density itself explains little of the variation ($r = 0.32$), which is what I expected: the knob is a way to perturb placement, not a physical cause. Ring capacitance spread and frequency spread move together across the whole set, from 4.7 fF and 4.19% at the tightest build to 7.8 fF and 6.99% at the widest.

Two details make the band trustworthy. The build at 60% density reproduces the shipped configuration exactly and returned 5.53%, matching the candidate build to the digit, and running the entire sweep a second time reproduced all nine values without change. The flow is deterministic, so the band measures genuine placement sensitivity rather than run-to-run noise. Against that band the older 10.5% build looks like a wide draw rather than a typical outcome. If I had run this sweep before writing up that build, I would not have quoted its number on its own.

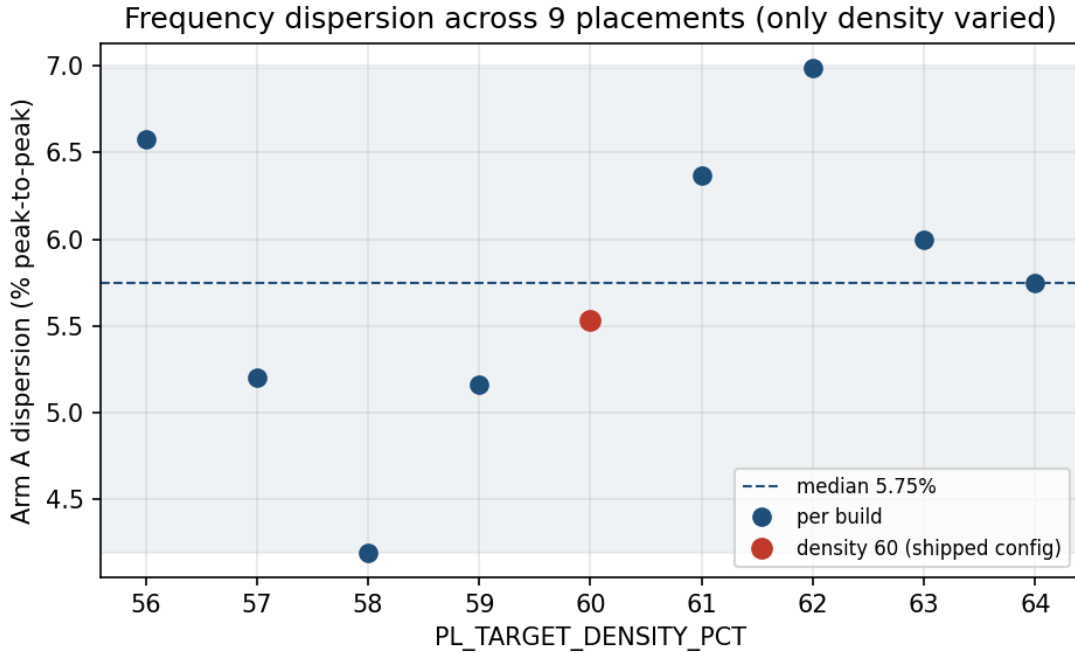


Figure 5. Nominal Arm A dispersion for nine builds that differ only in target placement density. The dashed line is the median, and the highlighted point is the shipped configuration.

5.4 The dispersion across process, voltage and temperature

Everything above is at 27 C and 1.8 V with typical devices, which bounds nothing. Repeating the same deck and the same extracted capacitances at a slow corner (100 C, 1.60 V) and a fast one (-40 C, 1.95 V) gives absolute frequencies of 276.2 to 291.7 MHz and 840.3 to 888.3 MHz, against 540.0 to 570.7 at nominal. All sixteen oscillators start at every corner, including the slow low-voltage one, and the no-parasitic control decks return a single identical frequency per corner (323.140, 633.640 and 987.948 MHz), which is what validates the corner setup.

The dispersion is essentially unchanged: 5.46%, 5.53% and 5.56% peak to peak at slow, nominal and fast. The frequency-capacitance correlation is -0.9997 in every case. The fitted slope scales with the operating frequency, -2.478, -4.936 and -7.783 MHz/fF, but expressed as a fraction of the mean it barely moves, -0.873, -0.890 and -0.902 percent per fF.

That is a stronger result than the nominal number alone. It says the routing contribution behaves as a relative perturbation of whatever speed the process and operating point set, rather than as a fixed frequency offset that a corner shift could swamp or amplify. For the silicon phase it predicts that the per-oscillator pattern should be recoverable from dies measured at different temperatures and supplies, which is convenient, because holding a hobby measurement setup at one temperature is difficult.

Two practical bounds fall out of the same simulations. The reported count is the oscillator frequency times the window duration, so at the fast corner the 16-bit counter reaches 35532 of 65535, leaving 1.84x headroom at the 25 MHz reference clock and the 1000-cycle

window in the design. Dropping the reference clock below 13.55 MHz, or extending the window past 1844 cycles, would push the fast corner into a silent wrap that returns a believable smaller count instead of an error.

5.5 Does the lumped model survive the real RC network?

Section 4 grounds each net's total capacitance on a single node. That discards the series resistance, collapses the split of capacitance between the ends of a net, and treats coupling as though the far end were held quiet. The last of those is the one to worry about here, because 47 to 62 of each ring's coupling capacitors have their far end on another node of the same ring, usually the neighbouring inverter. Adjacent inverter outputs move in antiphase, so grounding such a coupling understates the load it presents.

To test it I rebuilt all sixteen oscillators from the extraction's own network, with per-node capacitances, the series resistors as extracted, and node-to-node capacitors wherever both ends lie inside the oscillator, then simulated the lumped and distributed versions of each ring under otherwise identical conditions. The reconstruction accounts for the whole of every net's declared capacitance.

Every oscillator is slower under the fuller model, by 2.16% to 5.69%, and the size of the shift tracks the ring's extracted load ($r = -0.589$). Because the heavier rings lose the most, the dispersion widens rather than shrinking: 5.55% peak-to-peak becomes 7.60%. The simplification is therefore conservative with respect to the paper's central claim. It understates the layout contribution by roughly a third instead of manufacturing it, which is the opposite of the failure mode a reduced model is usually suspected of.

The per-oscillator pattern also survives. Rank correlation between the two models is 0.912 and the fastest ring is the same under both. The slowest label moves between two oscillators that the lumped model separated by 0.7%, which is a near-tie changing hands rather than the fingerprint dissolving.

Where the fuller model does change the answer is in the individual response bits. The design forms bits by comparing neighbouring oscillators, and two of the eight comparisons reverse. Both involved pairs separated by 0.28% and 0.32% under the lumped model, while every pair separated by 0.69% or more kept its ordering. So predicted bits from closely matched pairs are model-dependent and are not reported here as predictions; the analysis code already flags such pairs as low-margin when scoring measured silicon, and this gives an independent pre-silicon reason to expect which ones will be fragile.

Two limits on this check. The Arm B macro has a separate extraction and has not been redone this way, so its 569.5 MHz reference remains a lumped-model result. And the extraction itself is a reduced per-net network rather than a field solution, with coupling to nets outside a given oscillator still grounded, which ranges from four such nets on the lightest ring to seventy-two on the heaviest.

6. Matched-macro arm

The matched construction hardens one oscillator as a 60 x 40 micrometre macro. The macro layout passes the available DRC, LVS, antenna, and connectivity checks, and Arm B places 16 instances of it on a regular grid.

The extracted macro carries about 11.0 fF of total ring capacitance. One nominal post-layout simulation gives 569.5 MHz against a no-parasitic control of 633.15 MHz. The earlier Arm A capacitance fit predicts 570.2 MHz at that load, 0.12% away, a useful cross-check on the model. The macro result also lands within 0.35% of the earlier Arm A mean, so matching did not move the operating point.

The macro is faster than the typical routed ring, 569.5 MHz against an Arm A mean of 554.7 MHz, which is what the lighter load predicts. It is not faster than every Arm A ring. RO7 in the candidate build reaches 570.7 MHz because the router happened to give it 10.9 fF, slightly less than the macro carries. The gap is about 0.2%, below what this lumped-capacitance model resolves, and the macro also pays for the boundary buffers that hardening inserted and Arm A does not have. So the defensible statement is that matching the internal layout put the macro ahead of the average automatically routed oscillator, not ahead of all of them.

Figures 3 and 4 draw Arm B as a single horizontal reference line at 569.5 MHz, because the sixteen instances share one internal layout and there is only one extracted simulation behind it. By construction the internal layout contributes zero spread; fabricated Arm B instances will still differ through device mismatch, top-level routing, supply, and temperature. The simulation establishes only the internal-layout contribution under nominal device parameters. Total Arm B dispersion requires per-instance integration parasitics and fabricated-device measurements, and whether Arm B ends up with a smaller total spread than Arm A is the measurement the chip exists to make.

Arm A spreads; the matched macro is one line

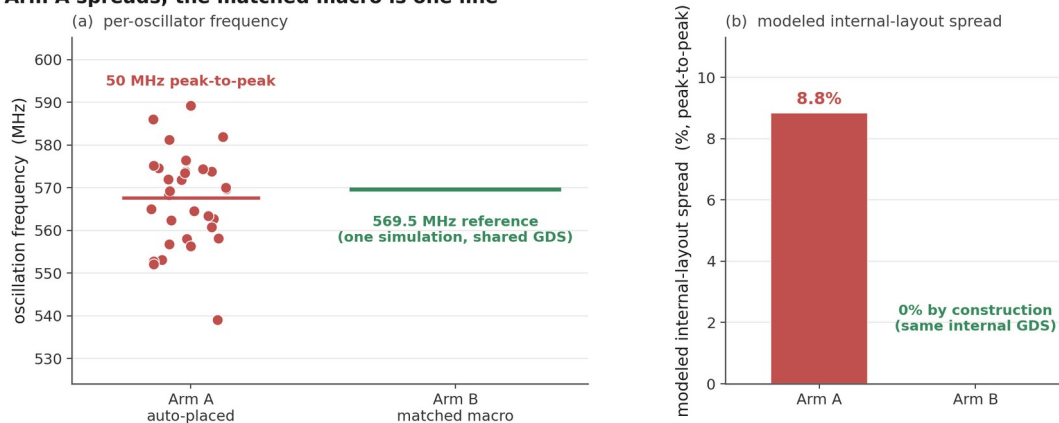


Figure 3. The earlier 32-oscillator array beside the matched-macro reference line at 569.5 MHz.

The dual-arm chip: Arm A spreads, Arm B is one layout

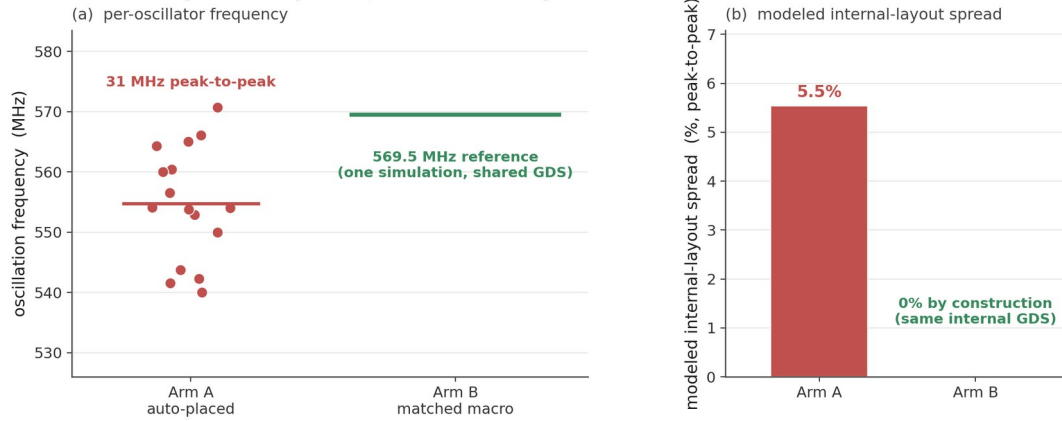


Figure 4. Arm A of the candidate build (5.53% peak to peak) beside the matched-macro reference line.

7. Planned silicon test

Why could a repeatable mask-defined pattern matter to a PUF at all? A layout component shared across dies can reduce uniqueness or make some comparisons predictable, but prior work also shows systematic structure is not automatically exploitable: Wilde, Hiller, and Pehl found adjacent-oscillator comparisons suppressed the spatial structure in their data and their predictor could not beat its baseline [4]. Which way this design falls is a question for fabricated dies, not for the nominal model. A toy population model and a first-order mismatch-scale estimate live in the repository's supplementary material (`sim/montecarlo.py`, `sim/spice/mc/`); their outputs depend on assumed amplitudes and a $\sqrt{31}$ scaling that the PDK's global mismatch draw does not really support, so no number from them appears here.

The question for silicon is whether Arm A retains more of its nominal layout pattern across dies than Arm B. The threat model I have in mind is concrete: an attacker knows the public design and mask and holds measurements from other dies of the same design, but none from the target die, and asks whether the shared deterministic layout component lets them predict the target die's pair ordering above the relevant per-bit baseline. Predictor accuracy is then judged against that baseline with whole chips held out, not against 50%. Testing it needs multiple chip IDs, repeated measurements, matched voltage and temperature settings, and a fixed comparison rule; the firmware records chip and condition labels so groups stay separate. The planned metrics are repeatability within chip and condition, centered pattern correlation across chips under the same condition, inter-chip Hamming distance for a predeclared set of adjacent comparison pairs, and within-chip response changes across voltage and temperature. Results count only when the grouping and completeness requirements are met; a single chip or an incomplete oscillator vector does not support a population claim.

8. Limitations

This study is pre-silicon, and its model is deliberately simple. Nominal transistor models carry no random local mismatch. The lumped-capacitance model has now been checked against the extraction's full RC network, as Section 5.5 reports: the dispersion survives and in fact grows, but individual comparisons between closely matched oscillators do not, and the Arm B macro has not been re-extracted that way. Neither model represents dynamic supply coupling between simultaneously active oscillators, which does not arise here because the hardware enables one at a time. The nine-build sweep is a controlled set in the sense that only one flow knob changed, but that knob is placement density and not the place-and-route seed, which LibreLane 3.0.3 does not expose. A seed sweep would be the cleaner experiment, and the band reported here should be read as placement sensitivity measured one particular way. Builds from earlier revisions are not part of that set and their spreads are not pooled with it. Instances within a single layout are also related observations, so instance-level confidence intervals would overstate the evidence. The Arm B reference is one simulation of one macro and cannot quantify fabricated Arm B variation. The 40-run global Monte Carlo cannot reproduce independent local device mismatch. Voltage, temperature, supply noise, ageing, package, and measurement-system effects are partly characterized: the corner simulations in Section 5.4 bound the frequency range and the counter margin, but supply noise, ageing, package effects and the measurement instrument itself remain unknown until chips exist. Uniqueness, reliability, min-entropy and attack success all need a multi-chip data set with a stated threat model. The corner work pairs device corners with nominal interconnect rather than pairing the slow corner with maximum extracted capacitance and the fast corner with minimum; device spread dominates the frequency bound, so the bound holds, but the range would tighten slightly under the fuller pairing. The boundary behaviour of the oscillator-clocked ripple counter is checked at both nominal and the fast corner, which is where the shorter period makes it hardest, though the selector path feeding it has not yet been validated as a whole chain at 888 MHz. None of this erases the modelled dispersion; it bounds what can be concluded from it.

9. Conclusion

Under a reduced lumped-capacitance model of the verified post-route topology, the automated flow gave the 16 oscillators of the candidate build a 5.53% peak-to-peak nominal frequency dispersion (population SD 1.65%, mean 554.7 MHz). Nine builds that differ only in placement density put that figure in context: median 5.75%, range 4.19% to 6.99%. Any single build reports one draw of the router, which is why I no longer quote one of them on its own. What repeats is the mechanism. Every build shows frequency tracking extracted ring capacitance at about -0.999, the fitted slope stays near -4.94 MHz/fF across independent layouts, and a fit trained on one build predicts another's individual frequencies to roughly 0.1%. A hardened macro provides a 569.5 MHz shared-internal-layout reference for Arm B, ahead of the Arm A mean though not of every Arm A instance. The finding is that the flow assigns materially different routing capacitance to logically identical oscillators, and that under nominal device assumptions this converts into a

dispersion of several percent through a relationship that holds from one routing run to the next.

Uniqueness, reliability, min-entropy, and attack success are not evaluated pre-silicon. Whether the mask-defined pattern survives fabrication, and how it compares against real device mismatch, will be settled by measuring both arms of the fabricated chips under the protocol in the firmware.

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